



LEGACY4REUSE'23



TOWARDS RIGOROUS DATA UPCYCLING USING THE FAIR IMPLEMENTATION PROFILE FOR THE SSHOC-NL SOCIO-ECONOMIC HISTORY COMMUNITY

Shuai Wang, Angelica Maineri



AUTHORS



Shuai Wang

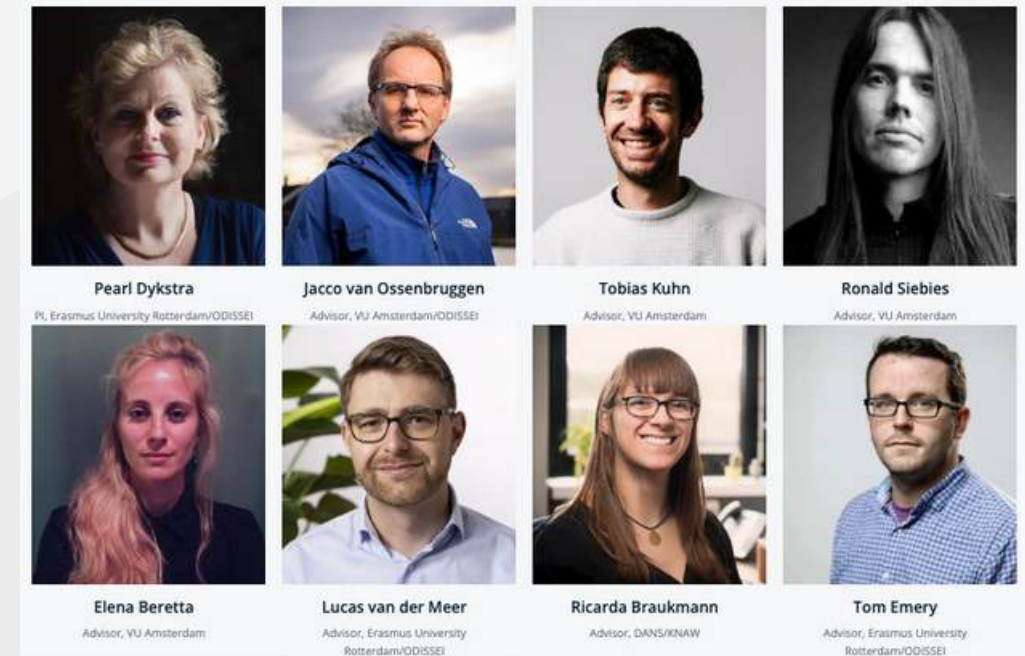
Scientific engiener
VU Amsterdam



Angelica Maineri

Data manager
ODISSEI &
Erasmus University
Rotterdam

Special thanks to



Rick Mourits
IISG



Tycho Hofstra
**University
Library, VU
Amsterdam**

<https://fairexpertisehub.org/>

TABLE OF CONTENTS

- Introduction to FIP
- SSHOC-NL-SEH FIP
- RQ & Methodology
- Legacy Datasets
- Upcycling using FIP
- Discussion
- Conclusion and future work



FAIR IMPLEMENTATION PROFILE

Definition

A list of decisions made by a community to FAIRify their data

The FIP is a collection of FAIR implementation choices made by a community of practice for each of the FAIR Principles.

FIC

A group of people who are using/publishing data on the same topics with some intention to conduct FAIR implementation

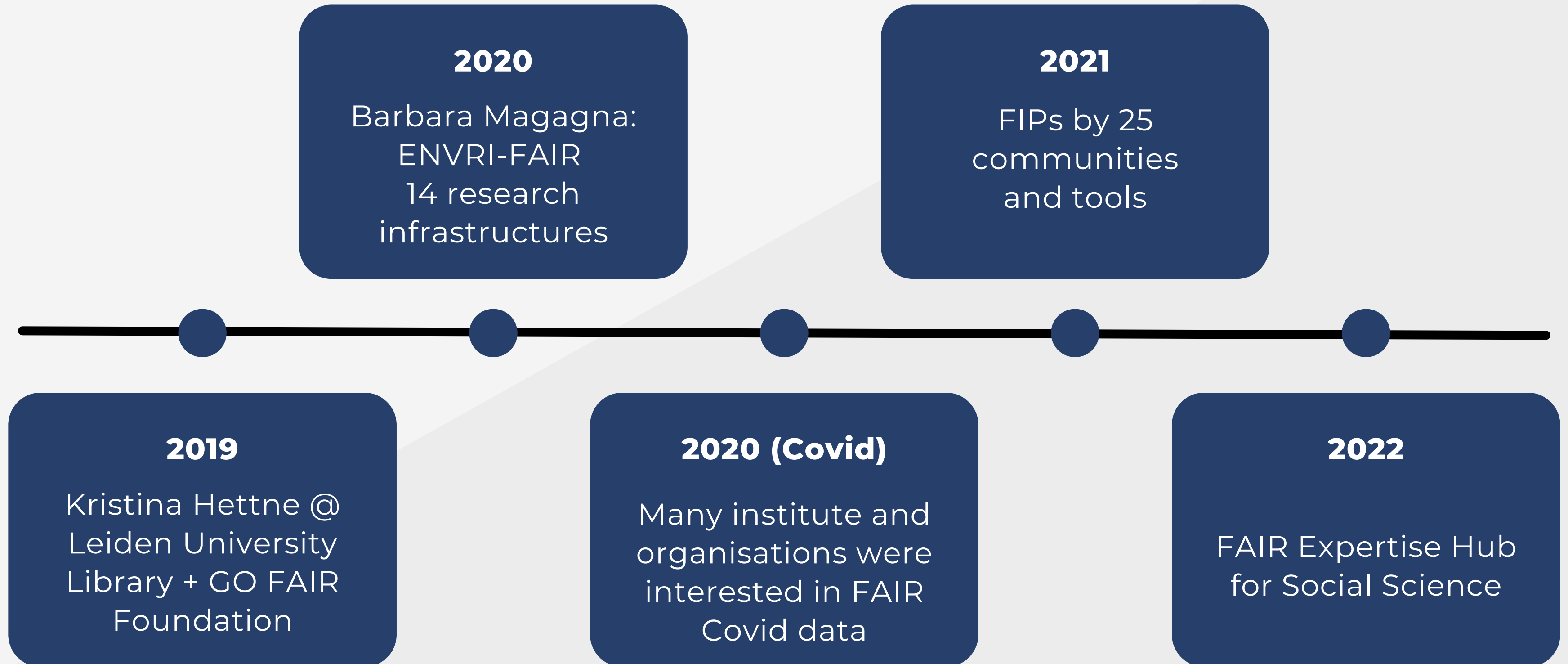
FER

A resource that enables the implementation of FAIR behaviors

Their relations

A FIC used a FER
A FIC is using a FER
A FIC will use a FER

A MINI HISTORY OF FIP



**VODAN
AFRICA**

HEALTH-RI

**ZONMW
COVID
PROGRAM**

**DUTCH
CANCER
FOUNDATION**

**DANISH
E-INFRASTRUCTURE
CORPORATION**

ENVRI-FAIR

**WORLD
FAIR**

**EOSC
NORDIC**

**NASEM
ROUNDTABLE ON
OPEN SCIENCE /
ORFG**

**NIAID / NIH
OFFICE OF
DATA
SCIENCE
STRATEGY**



- Two FIPs in WorldFAIR
- One FIP from past attempts
- Data in social science can be very different (e.g. more qualitative data)
- Diversity in data (bio, geospatial, images)
- Different data management behaviors
- Less FAIR

COMMUNITIES

ESS

European Social
Survey (ESS)

AUSSI-ESS

Australian Social
Survey International –
European Social Survey

GESIS SSSR

GESIS Social
Science Survey
Research

SEH

SSHOC-NL Socio-
Economic History

MCAL

Media Content
Analysis Lab
scholars

LGBTQVoc

LGBTQ+ Linked
Open Vocabulary

Nov 2022 - Apr 2023

SEH

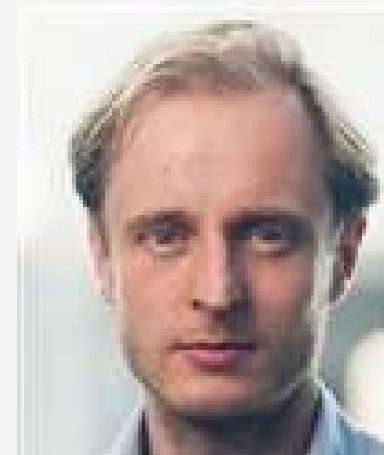
SSHOC-NL Socio-
Economic History



Richard Zijdeman
IISG



Rick Mourits
IISG



Auke Rijpma
Utrecht Universiteit



Xander Wilcke
Vrije Universiteit
Amsterdam



Sytze Van Herck
Utrecht Universiteit





Meetings

Processing

Review

Publish

CREATING FIPS

- Meetings/emails with members of the communities.
- All FIPs were created on the FIP Wizard
- Download in PDF and *.trig

LGBTQ+ Linked Open Vocabulary FIP

Questionnaire

Metrics

Preview

Documents

Settings

View

TODOs

Comments

Version history

Current Phase

Defining FAIR Imple...

Chapters

I. About

II. Declare your FAI...

III. Declarations for ...

IV. Declarations for ...

V. Declarations for I...

VI. Declarations for ...

VII. Register a new re...

In this section of the FIP Wizard, you, the Community Data Steward, are requested to answer questions that will declare, in a machine-readable way, the FAIR Implementation Community.

✓ II.1 Select your FAIR Implementation Community

In the drop-down list, select your FAIR Implementation Community. If your community is not yet listed, you will need to first create a nanopublication referencing your FAIR Implementation Community. Once you create this nanopublication, you will then see it in the drop-down list. A Wizard for creating nanopublications referencing a FAIR Implementation Community can be accessed below (in Chapter VII. Register a new resource as a nanopublication).

☒ Desirable: Defining FAIR Implementation Profile

LGBTQ+ Linked Open Vocabulary Community

The LGBTQ+ Linked Open Vocabulary Community consists of researcher, librarians, linguistics experts that are contributing to the development and use of vocabularies such as Homosaurus, QLIT, etc. The community is centered around the Homosaurus project.

- See more here

http://purl.org/np/RAOGUK-HzDHdTUPVG9tJXEjnG_p1mkqOyIWUi10xf11_8#LGBTQVoc

Clear answer

Answered 3 months ago by Shuai Wang.

✓ II.2 Who is the Community Data Steward?

As the Community Data Steward, you take the responsibility to represent your FAIR Implementation Community, and for reporting the decisions your community has made on how to implement the FAIR Principles. Provide your ORCID here (like this: 0000-0003-2, without https://orcid.org/).

THE SSHOC-NL SOCIO-ECONOMIC HISTORY FIP

- The identifier captured by the FIP are Handle and DOI (to be used in the future) for datasets.
- Dataverse and DANS SSH Data Station (to be used in the future) are the chosen search engines.
- XML, HTML, RDF are the knowledge representation languages captured in this FIP for its datasets.
- That for metadata includes RDF (OWL, RDFS), and SKOS.
- Finally, CC-BY-SA is being used by the community with CC-BY-NC to be used in the future.



ANALYSIS USING FIPS

Use of FAIR
Enabling
Resources

FIP Similarity
analysis

FAIR
Convergence
Analysis



FIP2DMP

Aligning DMPs with
community standards
using FIPs

FAIR IMPLEMENTATION PROFILE



FIP4Upcycling

Can FIPs provide
suggestions for the
upcycling of datasets?

Research question

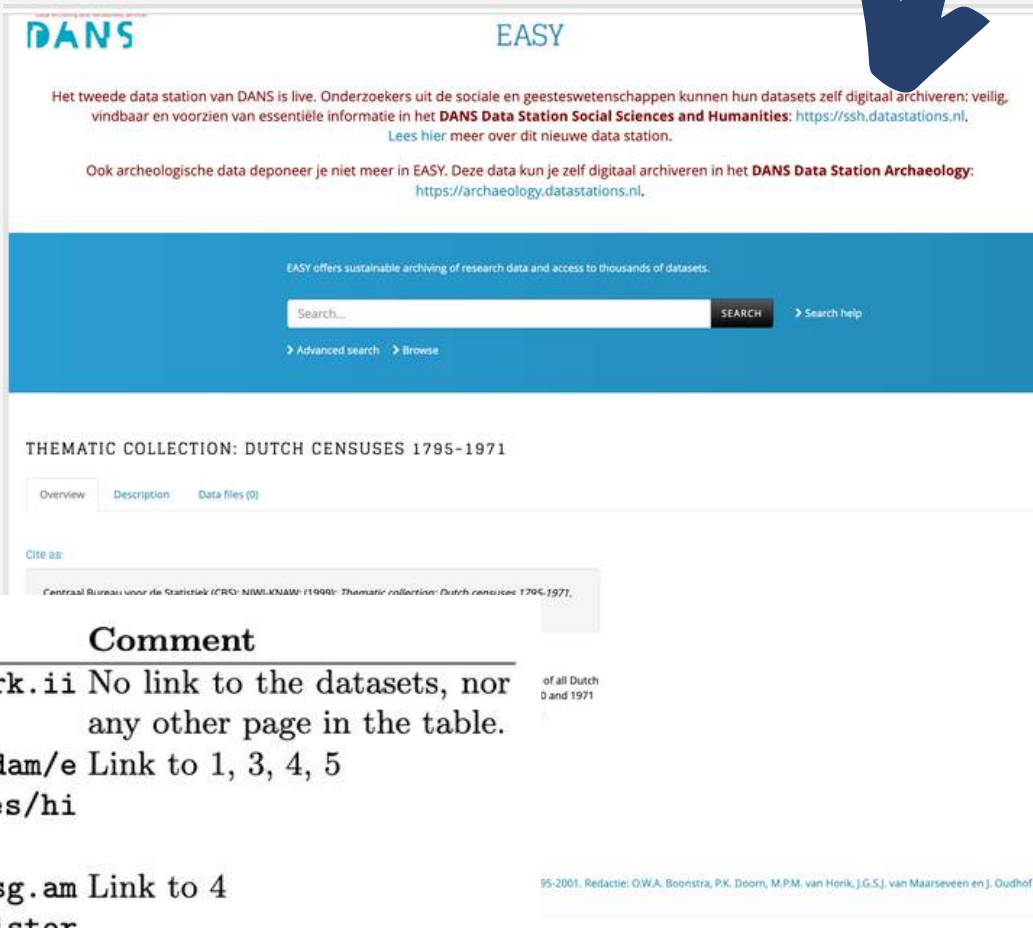
METHODOLOGY

1. Manual examination of the upcycling of some datasets
2. Create a table with all the entries under assessment
 - **F**indability: Persistent and resolvable identifier, search engine and repository.
 - **I**nteroperability: metadata format, metadata knowledge representation language, the format for datasets, knowledge representation for datasets.
 - **R**eusability: the licence of datasets.
3. Compare against the FIP

LEGACY DATA AND DARK DATA

Dataset	Year created	Year last maintained	Comment
Volkstellingen	1997	2007	Some census tables published between 1795 and 1971. (volkstellingen.nl)
Linked International Classification for Religions (LICR)	2017	2017	A classification system that provides mappings to various other well-known systems such as IPUMS, NAPP, HL7. (www.licr.io)
History Of Work Information System (HISCO)	2002	2020	A multilingual class scheme that provides historical occupational codes and classification of occupational activities
Gemeentegeschiedenis (Municipal History)	2007	2020	The boundaries of Dutch municipalities since 1812.
Historical Database of Dutch Municipalities (HDNG)	1970	2001	Historical Dutch Municipalities

- The official website (www.licr.io) is no longer accessible.
- Volkstellingen was published on EASY. But EASY is not maintained anymore
- Gemeentegeschiedenis has 1542 datasets published in the EASY repository but are being migrated to the DANS SSH Data Station
- The datasets of Gemeentegeschiedenis were converted to GeoJSON and maintained by IISG.
- HDNG: the dataset can pass through multiple people
- HISCO was published at different locations but no link redirecting to the latest place



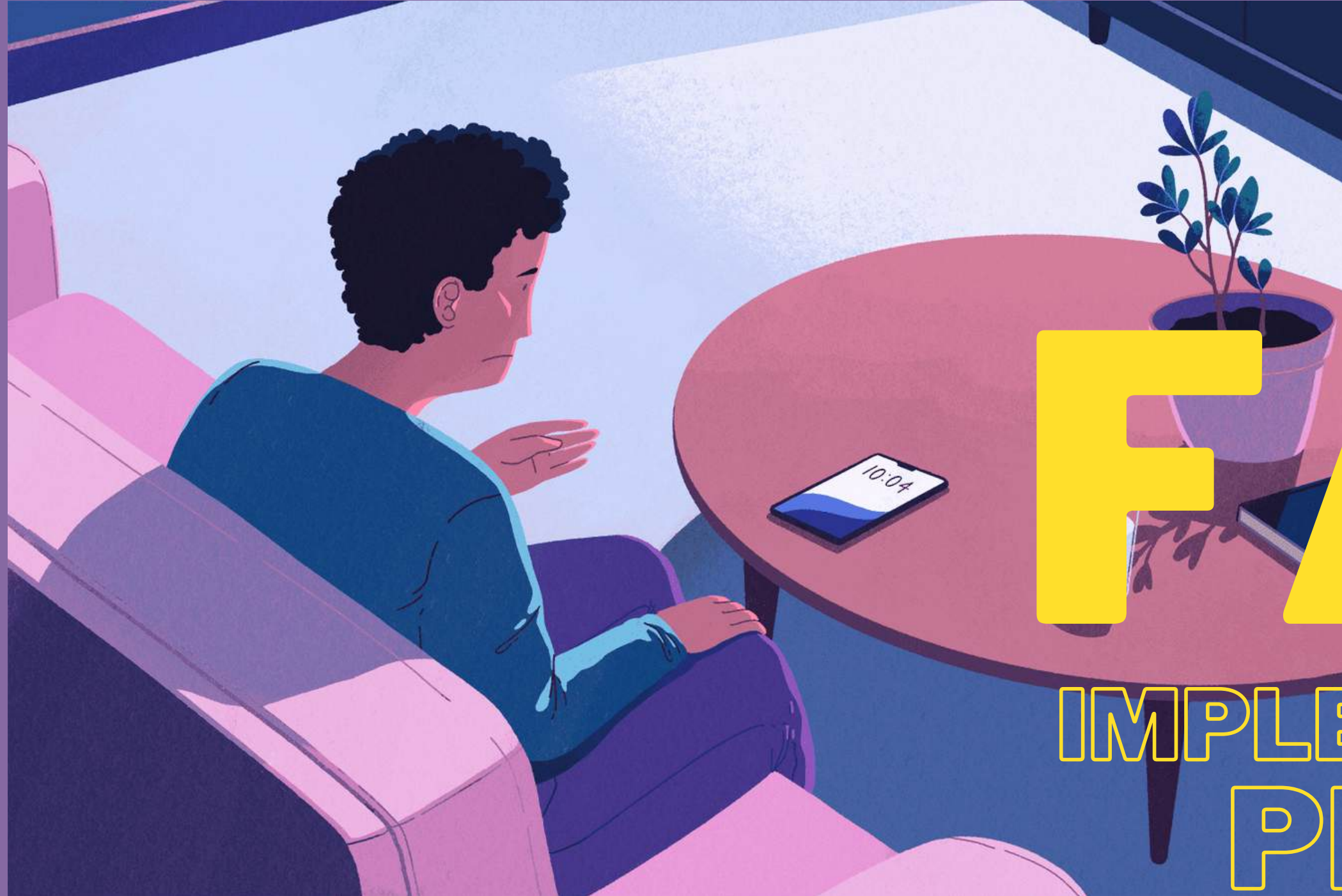
Version	Version History on Dataverse	Contributors	Date Published
4	5.1	Ineke Kellij	Jul 2023
	5.0	Rick Mourits, Richard Zijdemann	Jul 2021
	4.0	Rick Mourits	Jul 2021
3	3.0	Richard Zijdemann	Feb 2021
	2.1	Richard Zijdemann, Ineke Kellij	Dec 2020
	2.0	Richard Zijdemann	Jan 2020
2	1.1	Richard Zijdemann	Feb 2019
	1.0	Richard Zijdemann	Mar 2016

Table 1. The version history of the HDNG dataset on the Dataverse of IISG

Index	Page	URI	Comment
1	HISCO page	https://www.licr.io/yofwork.ii	No link to the datasets, nor any other page in the table.
2	IISG HISCO page	https://iisg.amsterdam/en/data/data-websites/history-of-work	Link to 1, 3, 4, 5
3	History Of Work Information System page on IISG Data-verse	https://datasets.iisg.amsterdam/dataverse/history-of-work	Link to 4
4	HISCO page on IISG Dataverse	https://datasets.iisg.amsterdam/dataverse/HISCO	
5	The HISCO collaboration website	https://datasets.iisg.amsterdam/	Link to 3.
6	The HSNDB Occurrences page on IISG Dataverse	https://datasets.iisg.amsterdam/dataset.xhtml?persistentId=hdl:10622/88ZXD8	The latest version of HISCO with the Persistent ID https://hdl.handle.net/10622/88ZXD8, which lands on IISG's page on Resources for HSNDB's Historical Sample of the Netherlands.

Table 3. Attempts for the upcycling of HISCO

SOMEONE TO TALK TO?



FAIR

IMPLEMENTATION
PROFILE

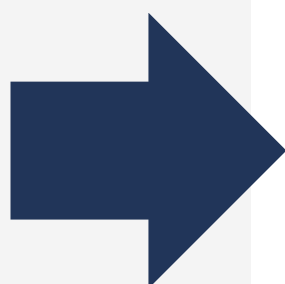
FOR OTHER COMMUNITIES TO
ALIGN TO

THE REFERENCE TO ALIGN
YOUR OWN LEGACY DATA

FAIR

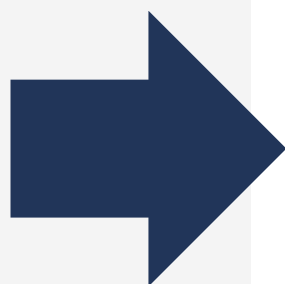
IMPLEMENTATION
PROFILE

Repo

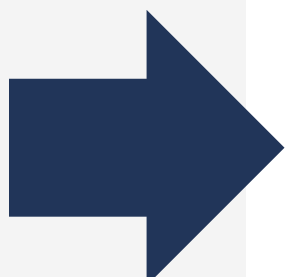


Aspects	LICR	Volkstellingen	HISCO	Gemeentegeshiedenis	HDNG	FIP
Year created	2017	1997	2002	2007	1970	-
Year last maintained	2017	2007	2020	2020	2021	-
Dataset PID	Handle	DOI, URN	Handle	DOI, Handle	Handle	DOI(to use), Handle
Repository	IISG Dataverse	EASY, DANS SSH Data Station	IISG Dataverse	DANS SSH Data Station, EASY, IISG Dataverse	Druid, IISG Dataverse	Dataverse, DANS SSH data station (future use)
Metadata format	Dublin Core, DDI, DataCite, DDI Codebook, JSON, OAI-ORE, OpenAIRE, Schema.org JSON-LD	XML, CSV	Dublin Core, DDI, DataCite, DDI Codebook, JSON, OAI-ORE, OpenAIRE, Schema.org JSON-LD	Dublin Core, DDI, DataCite, DDI Codebook, JSON, OAI-ORE, OpenAIRE, Schema.org JSON-LD	Dublin Core, DDI, DataCite, DDI Codebook, JSON, OAI-ORE, OpenAIRE, Schema.org JSON-LD	-
Knowledge representation language for metadata	XML, Dublin Core (DCMI Metadata Terms), DataCite, HTML, JSON-LD	Dublin Core (DCMI Metadata Terms), EASY Metadata Terms	XML, Dublin Core (DCMI Metadata Terms), DataCite, HTML, JSON-LD	XML, Dublin Core (DCMI Metadata Terms), DataCite, HTML, JSON-LD	XML, Dublin Core (DCMI Metadata Terms), DataCite, HTML, JSON-LD	RDF (OWL, RDFS), SKOS
Data format	NO, TAB	Various	CSV, TAB	Shapefile (.shp), dbf	RDF (Trig)	

KR



KR



Licence



	BOOK, JSON, OAI-ORE, OpenAIRE, Schema.or JSON-LD		BOOK, JSON, OAI-ORE, OpenAIRE, Schema.or JSON-LD	OpenAIRE, Schema.or JSON-LD	BOOK, JSON, OAI-ORE, OpenAIRE, Schema.or JSON-LD	
Knowledge represen- tation language for meta- data	XML, Dublin Core (DCMI Metadata Terms), DataCite, HTML, JSON-LD	Dublin Core (DCMI data Terms), EASY Meta- data Terms	XML, Dublin Core (DCMI Metadata Terms), DataCite, HTML, JSON-LD	XML, Dublin Core (DCMI Metadata Terms), Dat- aCite, HTML, JSON-LD	XML, Dublin Core (DCMI Metadata Terms), DataCite, HTML, JSON-LD	RDF (OWL, RDFS), SKOS
Data for- mat	NQ, TAB	Various	CSV, TAB, PDF, TXT	Shapefile (.dbf, .shp, .shx), Geo- JSON	RDF (Trig), CSV	-
Knowledge represen- tation lan- guage for datasets	SKOS, RDF, RDFS	Not found by manual exami- nation	None	None	RDF, SKOS	XML, HTML, RDF (RDFS, OWL)
Licence	CC0 1.0	Not found	CC0 1.0	CC BY-SA 4.0	CC0 1.0	CC-BY-SA, CC-BY-NC (new to the community)



DISCUSSION

- Upcycling is a continuous progress
- FIPs are a live document
- The FIP could have missed some entries
- FIP can help handle differences due to the dynamics of HR
- Not good at capturing resources beyond FER (yet)
- No automatic way to handle the upcycling process
- A dataset could be useful for multiple communities

Conclusion



Close examination of the upcycling of datasets in the SSHOC-NL Socio-Economic History community

Address problems in the upcycling of datasets

By mapping to questions of FIP, we explore the use of FIP as a community standard for the upcycling of datasets

Provide a list of suggestions for datasets

Discussed potentials of this approach and addressed problems and limits



FUTURE WORK

- The FIP could be updated
- Take more datasets for analysis
- Other communities (excluding social survey communities?)
- Accessibility could be included in some future work
- Aligning with other communities when upcycling





Do you think it worth further development?

THANK YOU

CONTACT

 shuai.wang@vu.nl

 www.fairexpertisehub.org



Platform Digitale Infrastructuur
Social Sciences & Humanities



ODISSEI



Clariah



References

- Erik Schultes et al. “Reusable FAIR Implementation Profiles as Accelerators of FAIR Convergence”.
- Mark D Wilkinson et al. “The FAIR Guiding Principles for scientific data management and stewardship”. In: Scientific data 3.1 (2016), pp. 1–9
 - Barbara Magagna et al. “FIPs and Practice”. In: Research Ideas and Outcomes 8 (2022). doi: 10.3897/rio.8.e94451.
 - Hana Pergl Sustkova et al. “FAIR Convergence Matrix: Optimizing the Reuse of Existing FAIR-Related Resources”. In: Data Intelligence 2.1-2 (Jan. 2020), pp. 158–170. issn: 2641-435X
 - Kristina Hettne et al. “FIP2DMP: Linking data management plans with FAIR implementation profiles”.

References (useful links)

FAIR Implementation Profile: 3-part video series:

<https://www.go-fair.org/2021/08/02/fair-implementation-profile-3-part-video-series/>

WorldFAIR: <https://docs.google.com/presentation/d/1vDgMCWW-RM806RnDul5ataGh4md5VX49u5NjcolVMU8/edit?usp=sharing>

FIP questionnaire and FER types: <https://bit.ly/WF-FIPs-Questions>

The FIP Wizard: <https://fip-wizard.ds-wizard.org/>

Image on slide 14: [https://www.google.com/url?](https://www.google.com/url?sa=i&url=https%3A%2F%2Fwww.verywellmind.com%2Fwhat-to-do-when-you-need-someone-to-talk-to-5089236&psig=AOvVaw1wTGhIERzSkO7mAnJ7HlvT&ust=1700596573423000&source=images&cd=vfe&opi=89978449&ved=OCBIQjRxqFwoTCJDSjliu04IDFQAAAAAdAAAAABAD)

[sa=i&url=https%3A%2F%2Fwww.verywellmind.com%2Fwhat-to-do-when-you-need-someone-to-talk-to-5089236&psig=AOvVaw1wTGhIERzSkO7mAnJ7HlvT&ust=1700596573423000&source=images&cd=vfe&opi=89978449&ved=OCBIQjRxqFwoTCJDSjliu04IDFQAAAAAdAAAAABAD](https://www.google.com/url?sa=i&url=https%3A%2F%2Fwww.verywellmind.com%2Fwhat-to-do-when-you-need-someone-to-talk-to-5089236&psig=AOvVaw1wTGhIERzSkO7mAnJ7HlvT&ust=1700596573423000&source=images&cd=vfe&opi=89978449&ved=OCBIQjRxqFwoTCJDSjliu04IDFQAAAAAdAAAAABAD)

Image on the conclusion page :[https://www.google.com/url?](https://www.google.com/url?sa=i&url=https%3A%2F%2Fstock.adobe.com%2Fsearch%3Fk%3Dconclusion&psig=AOvVaw0H1eSQLGq4HIE485Br-Xxu&ust=1700644008920000&source=images&cd=vfe&opi=89978449&ved=OCBIQjRxqFwoTCMCu--Le1IIDFQAAAAAdAAAAABAD)

[sa=i&url=https%3A%2F%2Fstock.adobe.com%2Fsearch%3Fk%3Dconclusion&psig=AOvVaw0H1eSQLGq4HIE485Br-Xxu&ust=1700644008920000&source=images&cd=vfe&opi=89978449&ved=OCBIQjRxqFwoTCMCu--Le1IIDFQAAAAAdAAAAABAD](https://www.google.com/url?sa=i&url=https%3A%2F%2Fstock.adobe.com%2Fsearch%3Fk%3Dconclusion&psig=AOvVaw0H1eSQLGq4HIE485Br-Xxu&ust=1700644008920000&source=images&cd=vfe&opi=89978449&ved=OCBIQjRxqFwoTCMCu--Le1IIDFQAAAAAdAAAAABAD)